Climate Data Melding via Spatially Varying Autoencoder

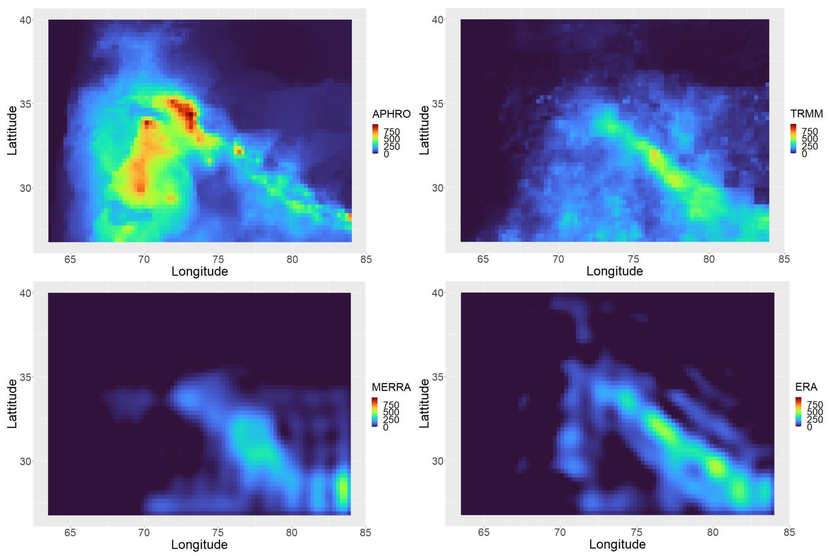
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Introduction

Glaciers and the precipitation they receive are important components of the hydrological cycle in High Mountain Asia (HMA). Our goal is to meld climate data from multiple climate data sources to create a consensus representation of precipitation levels in HMA.

Data

Our data comes from four different sources: TRMM, APHRODITE, MERRA, & ERA. Each climate data product uniquely represents the amount of precipitation that the HMA area received in July 2015, as seen below.

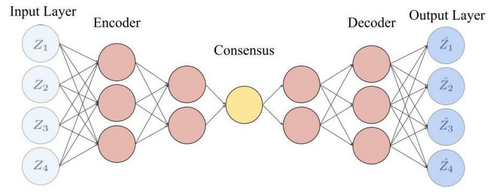


The amount of precipitation modeled for July 2015 by the 4 climate models

Our goal is to capture the unique features of each data product in order to model and understand the underlying spatial structure of rainfall by creating a consensus.

Autoencoder Structure

To meld the data from each climate model, we created a spatially varying Bayesian autoencoder using adaptive MCMC. The structure and underlying mathematics of the autoencoder are described in the figures below. Each neuron of the input layer represents the result from a specific climate model, which is then encoded to reach the consensus. We added constraints on the decoder weights to ensure model identifiability and interpretability. The weights are created using spatial basis functions with coefficients that spatially smooth the weights at each spatial location and each layer.



$$Z_p^{(l)}(s) = a\left(\sum_{j=1}^{P_{l-1}} Z_j^{(l-1)}(s) \quad W_{p,j}^{(l)}(s)\right)$$

$$W_{P,j}^{(l)}(s) = b'(s) \;\; \Theta_{P,j}^{(l)}$$

$$\boldsymbol{W^{(5)}}(s) = \begin{pmatrix} 1 & W_{2,1}^{(5)}(s) \end{pmatrix}$$

$$\boldsymbol{W^{(6)}}(s) = \begin{pmatrix} 1 & W_{2,1}^{(6)}(s) & W_{3,1}^{(6)}(s) \\ 0 & 1 & W_{3,2}^{(6)}(s) \end{pmatrix}$$

$$\boldsymbol{W^{(7)}}(s) = \begin{pmatrix} 1 & W_{2,1}^{(7)}(s) & W_{3,1}^{(7)}(s) & W_{4,1}^{(7)}(s) \\ 0 & 1 & W_{3,2}^{(7)}(s) & W_{4,2}^{(7)}(s) \\ 0 & 0 & 1 & W_{4,3}^{(7)}(s) \end{pmatrix}$$

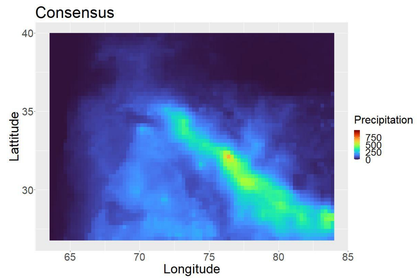
Model Tuning

While finding the best consensus, we tuned the following aspects of the autoencoder:

- The number of hidden layers
- The number of neurons in each hidden layer
- Which climate models is constrained
- The number of basis functions

From these combinations of tuning parameters, we have selected the best performing consensus.

Results



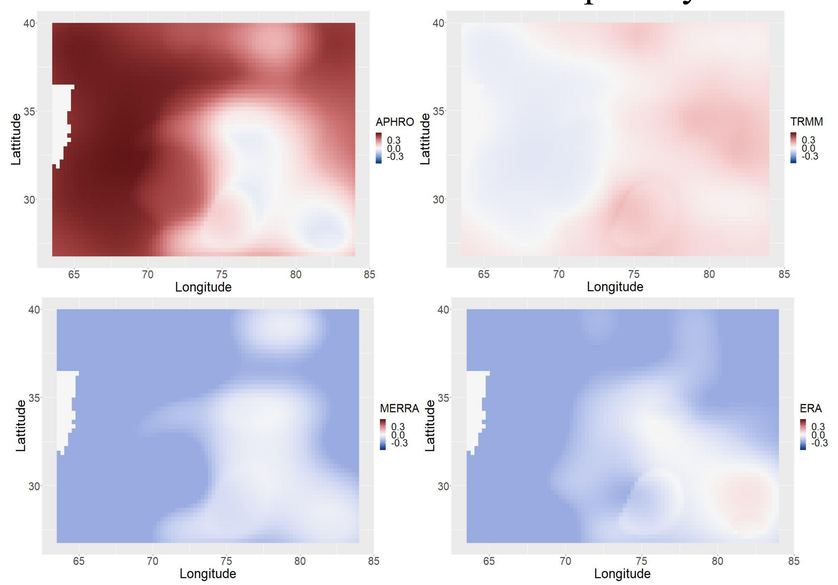
Consensus for July 2015

After tuning, we were able to recreate the spatial restructure of the consensus, which contains features from all four of the original climate models.

To test the value of the consensus, we calculated R-squared values of how well the consensus predicts the input data without being decoded.

APHRO	TRMM	MERRA	ERA
0.4009199	0.8973174	0.6713491	0.7379255

Lastly, we compared how the contribution of each climate model's data varies spatially.



Consensus Contributions by each Data Product

We see that each data product adds a unique contribution to the consensus; giving evidence that a spatially varying model captures more information than a normal average.